Convolution to Cryptography (conv2crypt)

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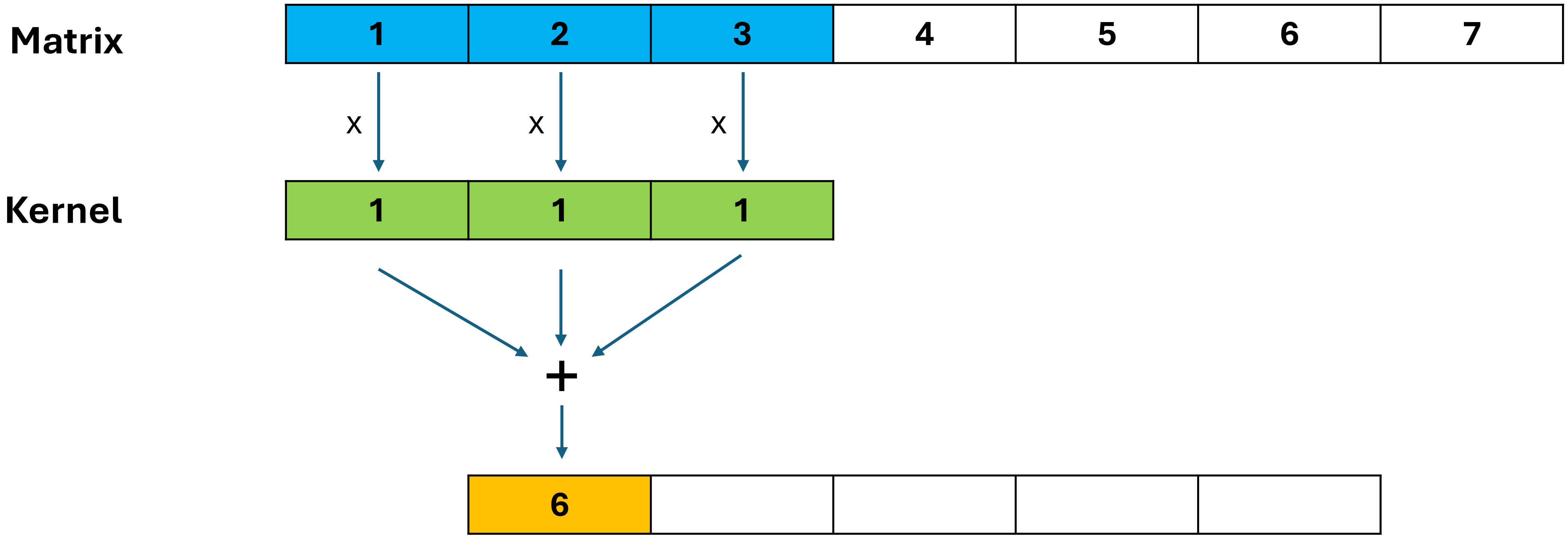
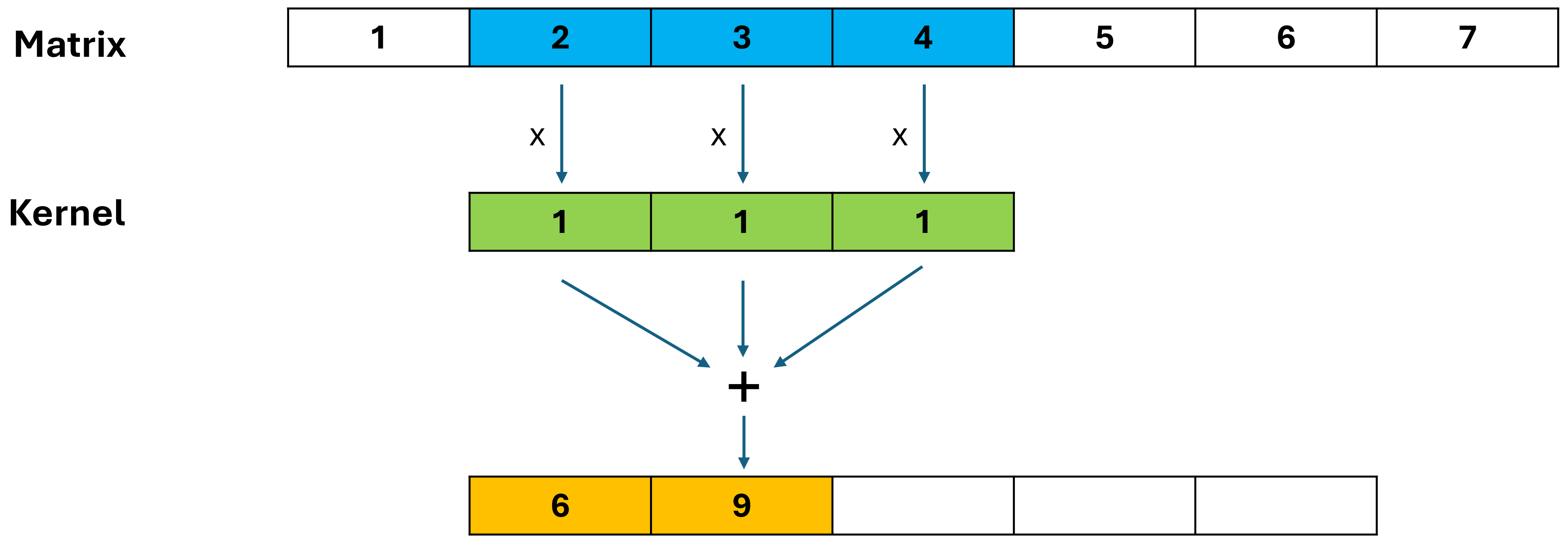
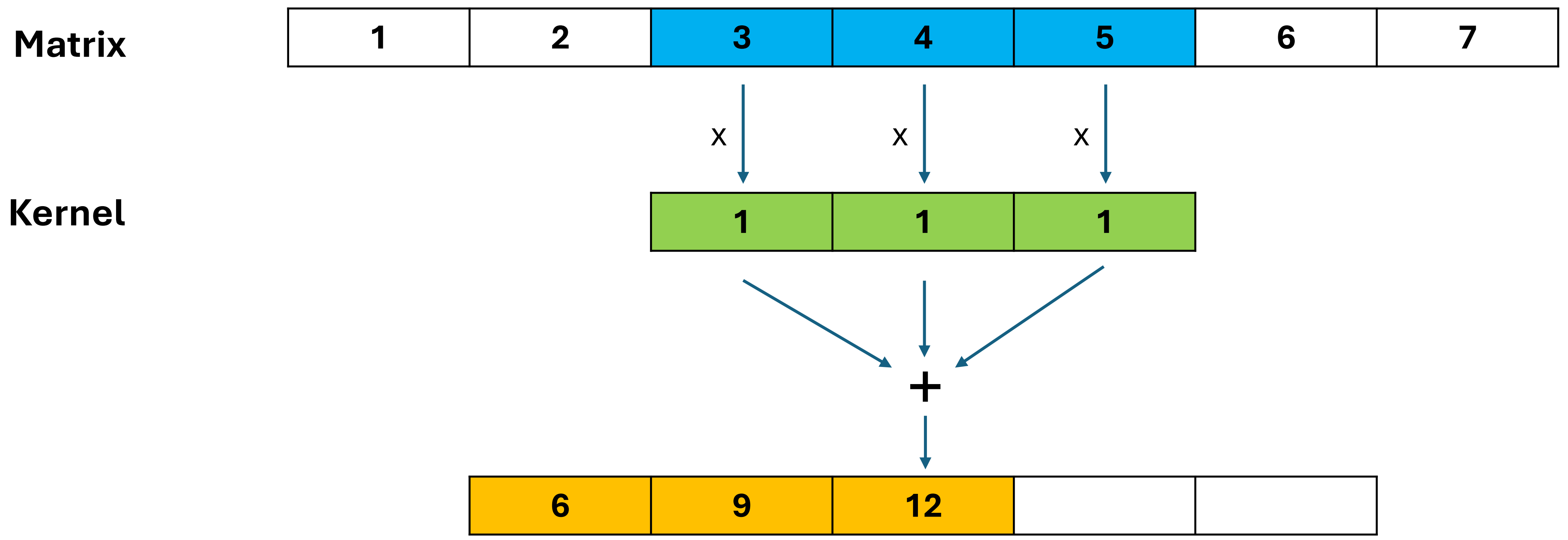
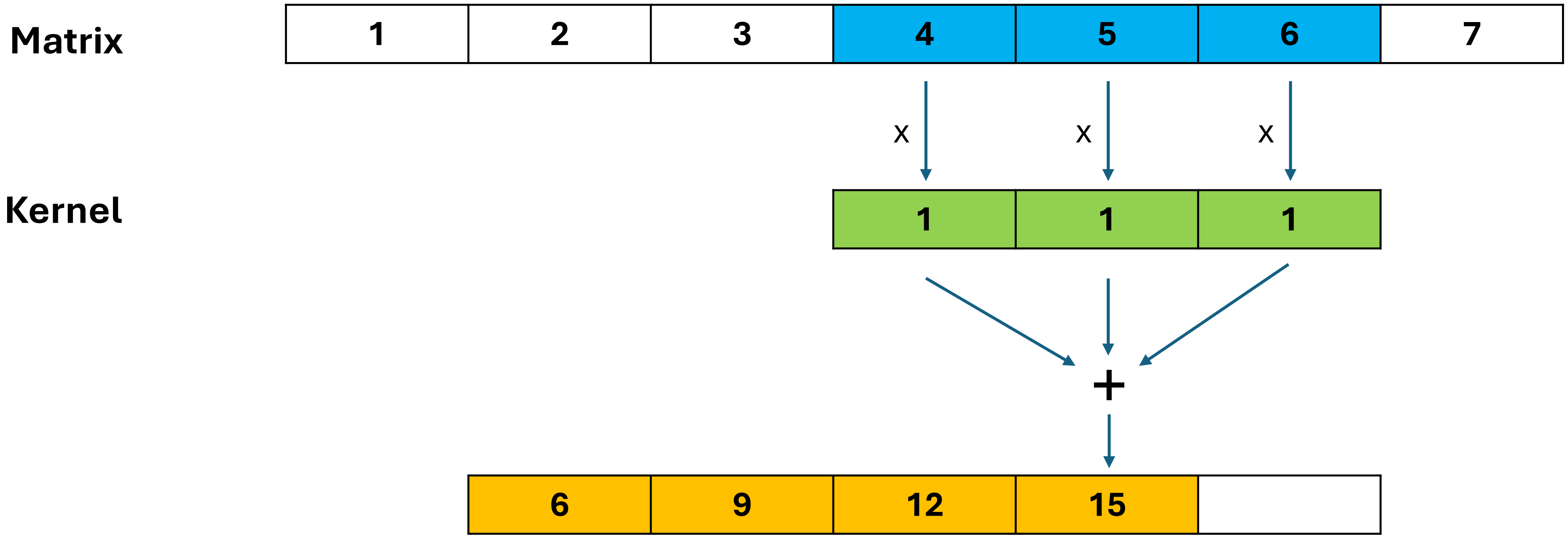
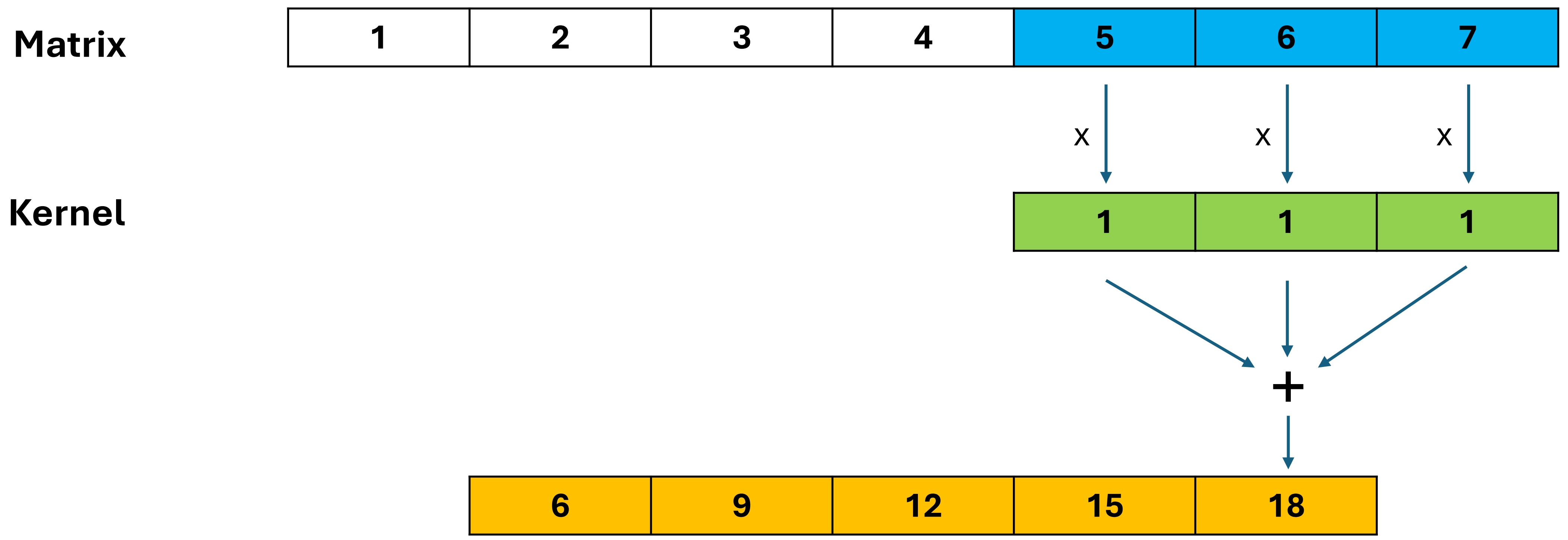

Matrix
1
2
3
4
5
6
7
x
x
x
Kernel
1
1
1
+
6

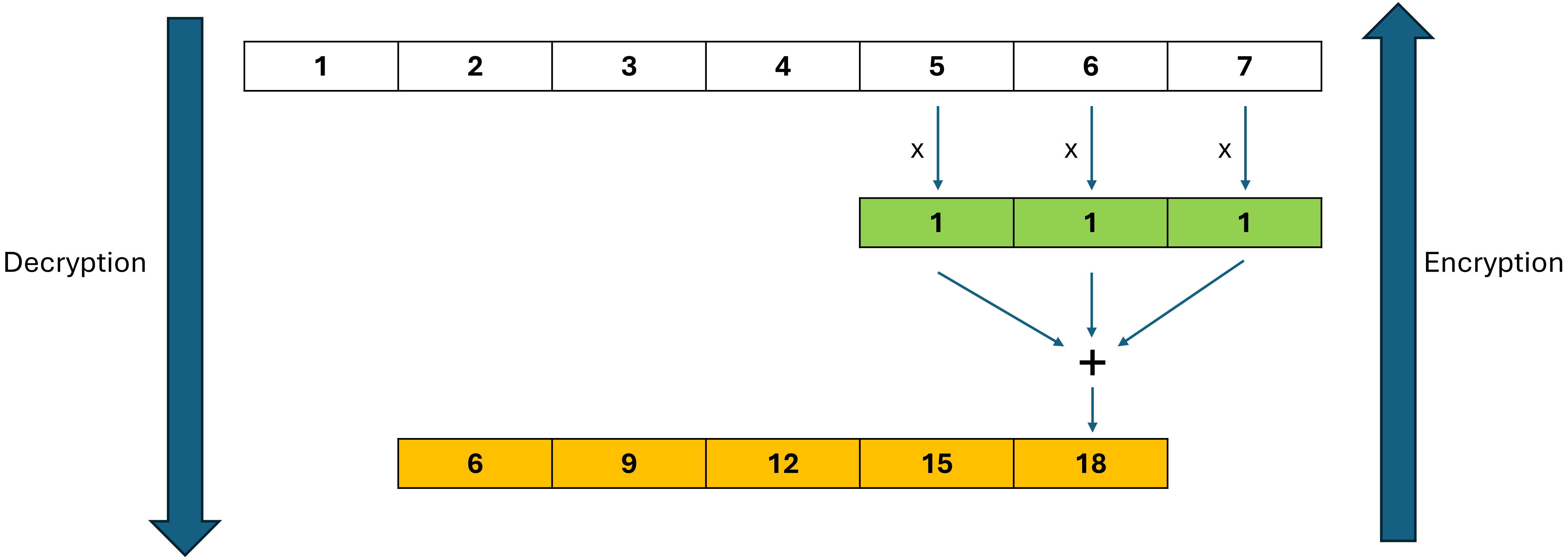


Matrix
1
2
3
4
5
6
7
x
x
x
Kernel
1
1
1
+
6
9

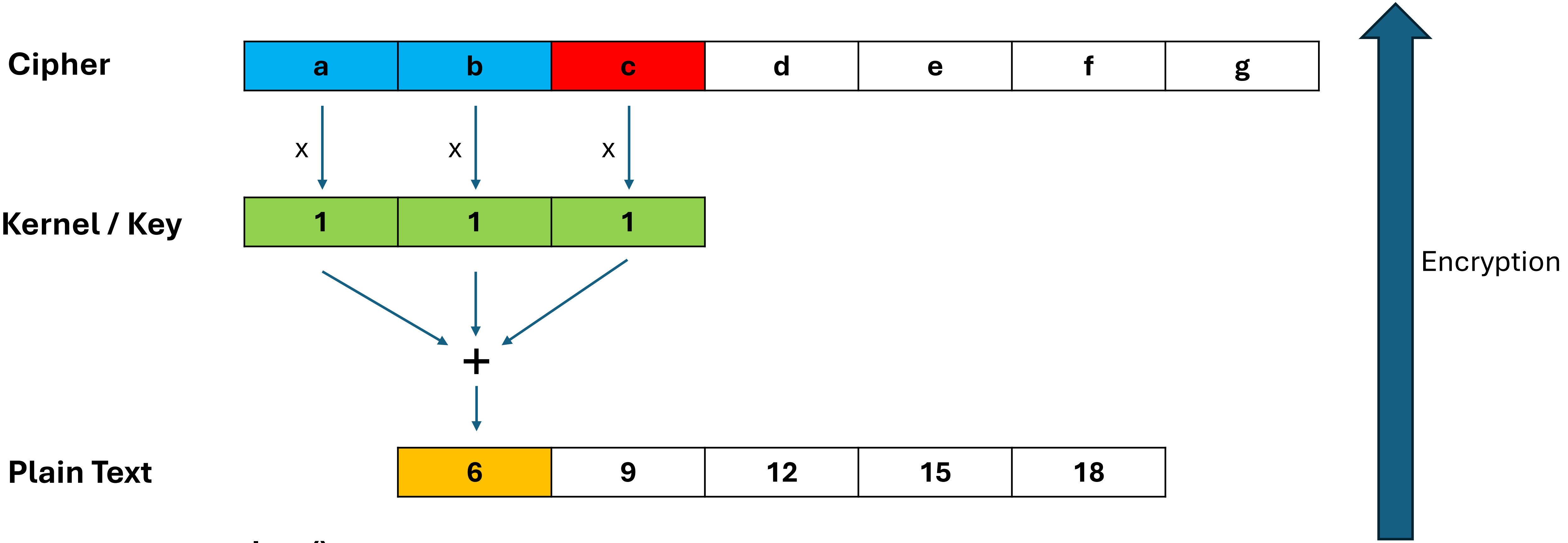

Matrix
1
2
3
4
5
6
7
x
x
x
Kernel
1
1
1
+
6
9
12


Matrix
1
2
3
4
5
6
7
x
x
x
Kernel
1
1
1
+
6
9
12
15


Matrix
1
2
3
4
5
6
7
x
x
x
Kernel
1
1
1
+
6
9
12
15
18



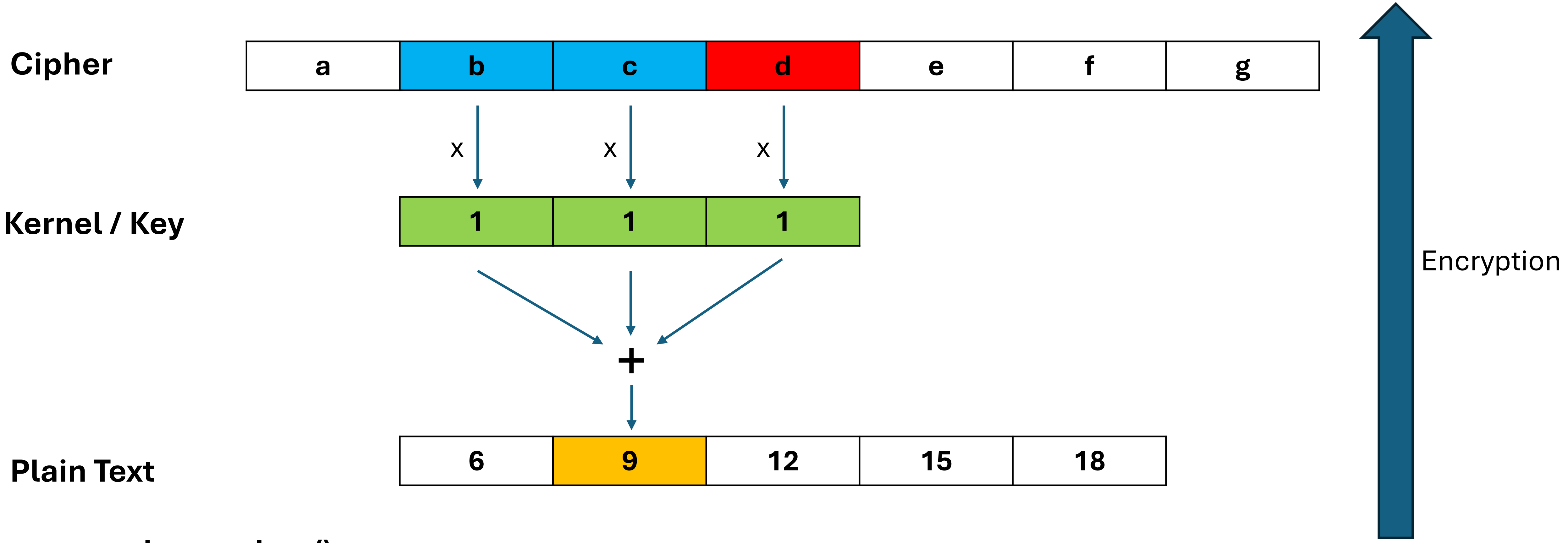
Decryption
Encryption
1
2
3
4
5
6
7
x
x
x
1
1
1
+
6
9
12
15
18



$a = \text{random}()$

$b = \text{random}()$

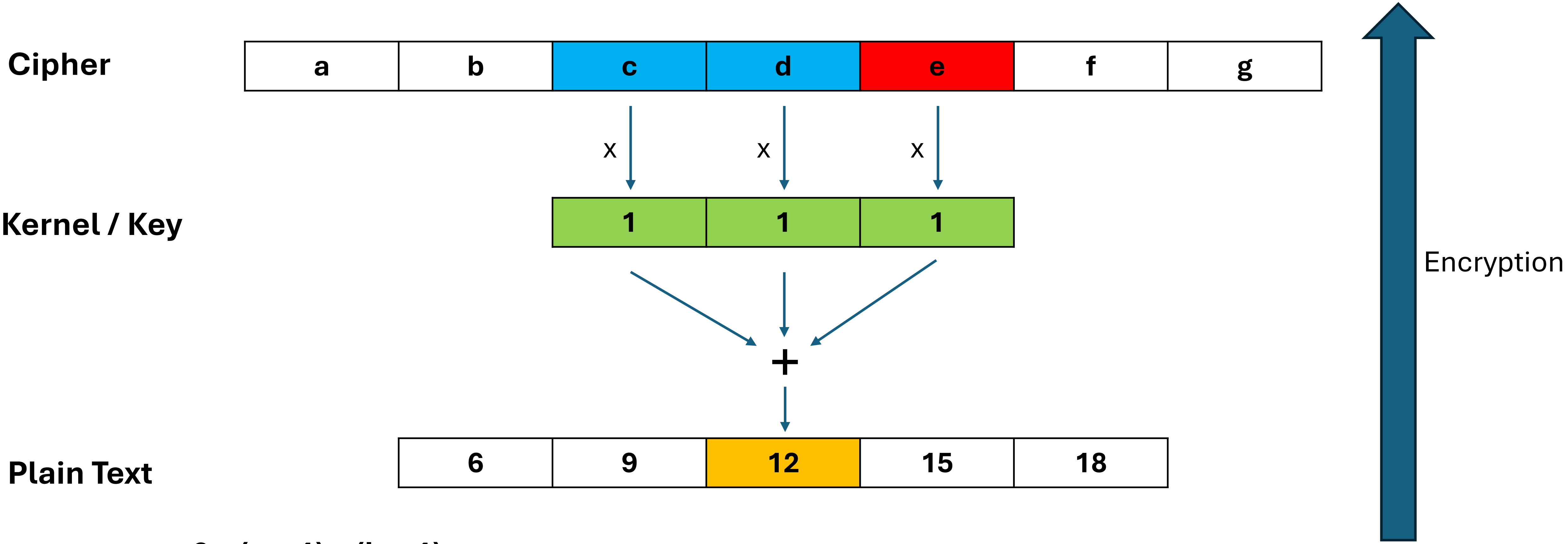
$6 = (a \times 1) + (b \times 1) + (c \times 1) \rightarrow \quad c = 6 - (a \times 1) - (b \times 1)$



$b = \text{random}()$

$c = 6 - (a \times 1) - (b \times 1)$

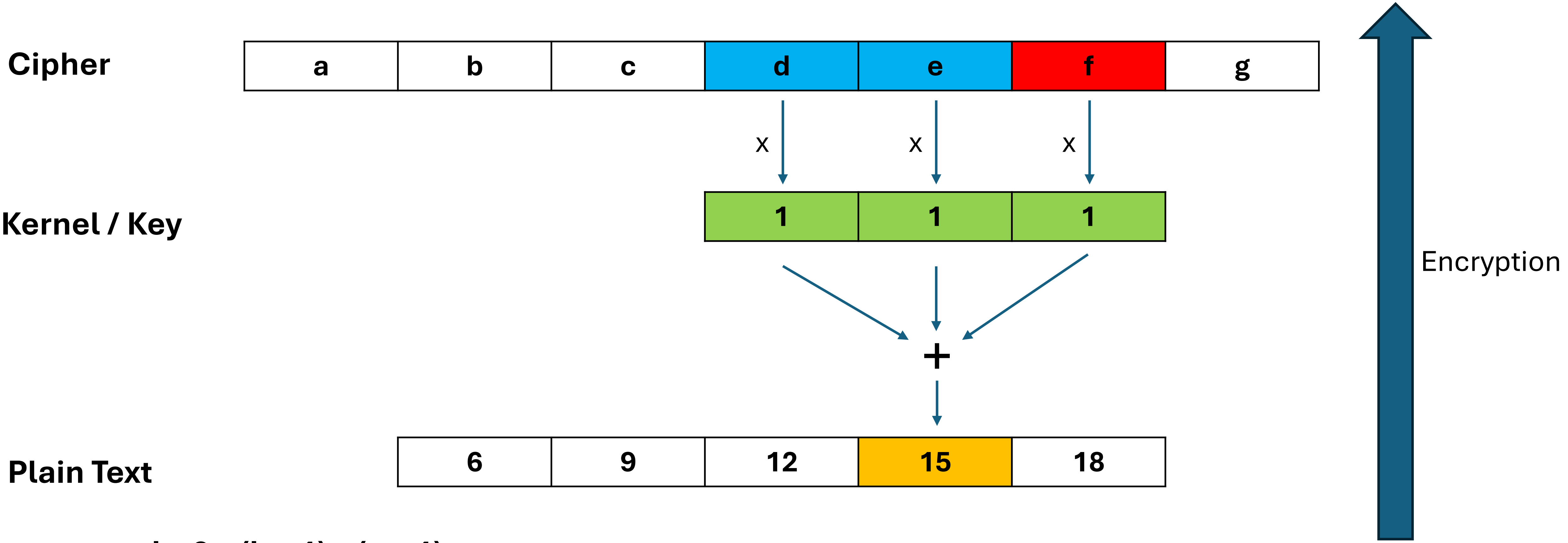
$9 = (b \times 1) + (c \times 1) + (d \times 1) \rightarrow \quad d = 9 - (b \times 1) - (c \times 1)$



$c = 6 – (a \times 1) – (b \times 1)$

$d = 9 – (b \times 1) – (c \times 1)$

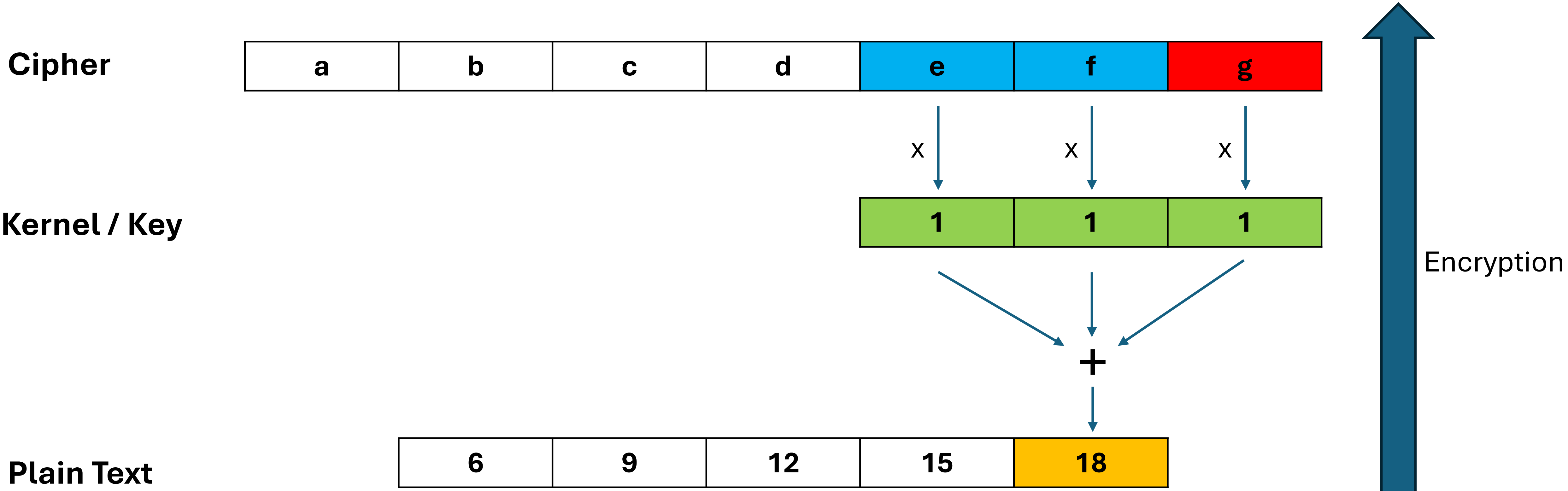
$12 = (c \times 1) + (d \times 1) + (e \times 1) \rightarrow \quad e = 12 – (c \times 1) – (d \times 1)$



$$d = 9 - (b \times 1) - (c \times 1)$$

$$e = 12 - (c \times 1) - (d \times 1)$$

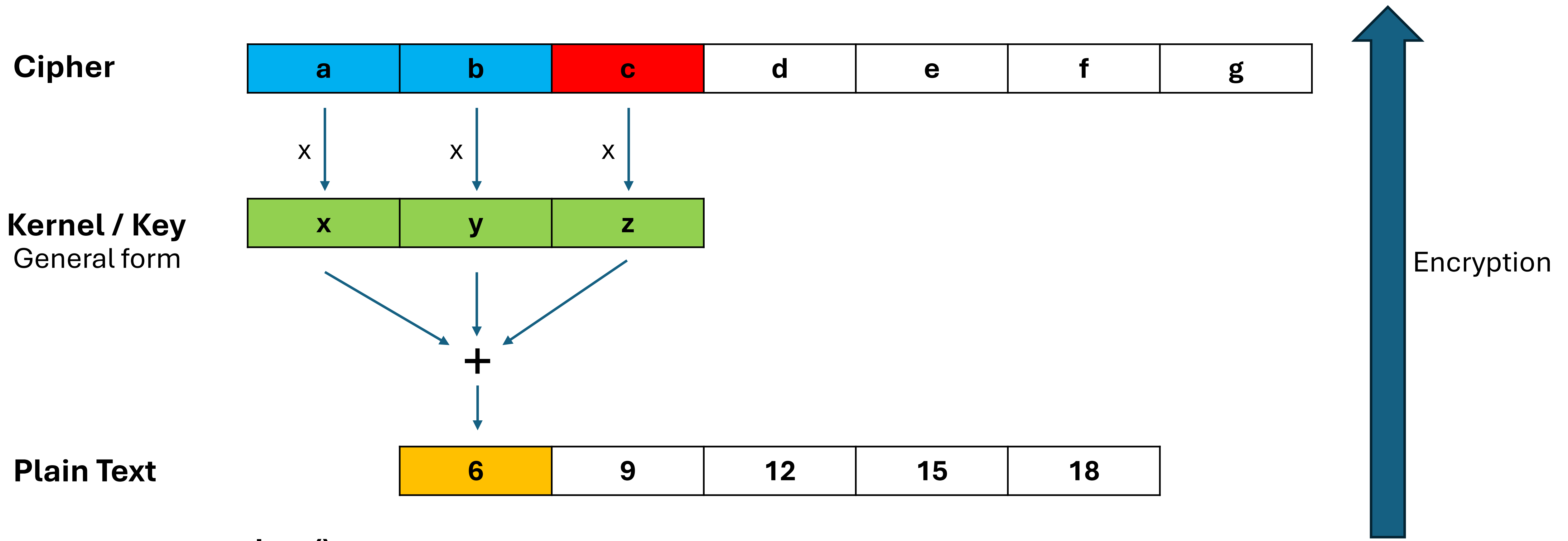
$$15 = (d \times 1) + (e \times 1) + (f \times 1) \rightarrow \quad f = 15 - (d \times 1) - (e \times 1)$$



$e = 12 - (c \times 1) - (d \times 1)$

$f = 15 - (d \times 1) - (e \times 1)$

$18 = (e \times 1) + (f \times 1) + (g \times 1) \rightarrow \quad g = 18 - (e \times 1) - (f \times 1)$



$a = \text{random}()$

$b = \text{random}()$

$6 = (a \times x) + (b \times y) + (c \times z) \rightarrow c = (6 - (a \times x) - (b \times y)) / z$

$$\boxed{z \neq 0}$$

** The only RULE for conv2crypt , the last index of the kernel / key must not be zero.